Milestone 7: Claims-Validated ROI Framework

Stars/QBP valuation range on research-generated lift

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RESEARCH FRAMEWORK. Lift input: Research-generated IncentiveRewards mock. Not InComm-issued lift, not InComm financials. Dollar values are research ranges tagged by provenance.

1. What the lift evidence says

- Lift status: LIFT_NOT_DISTINGUISHABLE_FROM_ZERO.
- ATE 0.0114 (SE 0.0187); minimum detectable effect 0.0524.
- ROI is reported only when lift is distinguishable from zero. A break-even is a floor, not an ROI.

2. Required lift for break-even

Independent of measured lift: the average lift per treated member at which the program pays for itself, under low/base/high value of one incremental closure.

Fraction Treated	Treated Rows	Reward Cost Expected Usd	Required Lift Pp For Breakeven Low	Required Lift Pp For Breakeven Base	Required Lift Pp For Breakeven High
0.1000	1167	12,805	16.9379	10.7947	6.4726
0.2500	2917	29,667	13.2600	8.4785	5.0867
0.5000	5834	54,580	10.2567	6.6040	3.9740
1.0000	11669	100,588	8.4423	5.4856	3.3151

3. Redemption sensitivity

Redemption is not measured. 0.988 is an OTC-benefit activation rate, not a reward redemption rate, so it is used only as the cost ceiling (every earned reward is paid). 0.75 and 0.55 are equally unsourced sensitivities. Cost, and so the break-even lift, scales with this rate: read the headline as the range below, not a point. A low redemption rate probably also means a smaller lift, because an unredeemed reward motivates less.

Redeem P	Fraction Treated	Reward Cost Expected Usd	Required Lift Pp For Breakeven Base
0.9880	0.1000	12,805	10.7947

0.9880	0.2500	29,667	8.4785
0.9880	0.5000	54,580	6.6040
0.9880	1.0000	100,588	5.4856
0.5500	0.1000	7,128	6.0092
0.5500	0.2500	16,515	4.7198
0.5500	0.5000	30,383	3.6763
0.5500	1.0000	55,996	3.0537
0.7500	0.1000	9,720	8.1944
0.7500	0.2500	22,520	6.4361
0.7500	0.5000	41,432	5.0132
0.7500	1.0000	76,358	4.1642

4. Which assumption matters most (tornado)

USD per incremental closure (mean over the treated gap mix) when one input moves from its value-lowering to its value-raising end and the others stay at base. Ranked by swing.

Assumption	Provenance	Value At Low Input	Value At High Input	Swing Usd
eligible_share (diabetes / hypertension)	Public source (CMS/KFF)	131.4515	200.8712	69.4197
qbp_per_enrollee_year	Public source (CMS/KFF)	131.1576	192.1995	61.0419
rho_boundary_density (CMS 2026 data)	Public data, computed (CMS 2026 Star Ratings)	146.3134	166.3828	20.0694

5. Uncertainty over lift and assumptions (Monte Carlo)

Population lift is drawn around the average treatment effect with its standard error (not the in-sample ranking gain, which is optimistic under selection) and each assumption uniformly between its bounds. These probabilities describe uncertainty in research inputs, not a forecast of InComm results.

Fraction Treated	P Net Positive	Net P05 Usd	Net P50 Usd	Net P95 Usd
0.1000	0.0047	-16,593	-10,694	-4,455
0.2500	0.0082	-39,135	-24,391	-8,797
0.5000	0.0144	-73,515	-44,029	-12,839
1.0000	0.0220	-138,463	-79,485	-17,101

6. Closure referential-integrity check (not an independent validation)

Referential-integrity check on InComm's export (claim ids MOCK-CLM-*). Closure is derived from the same claim, so member, CPT, date and window agree by construction; 100% shows internal consistency, not clinical validity or production data quality. Gaps sharing a claim map to the same HEDIS measure (e.g. DIAB_A1C_TEST and DIAB_BLOOD_SUGAR both -> CDC-A1C), so measure value is de-duplicated across them.

Care Gap Code	Closed Gaps	Claim Found	Same Member	Cpt In Definition	Date Matches	In Window	Gaps Sharing A Claim
BH_POST_DISCHARGE_7D	1248	1248	1248	1248	1248	1248	0
COPD_SPIROMETRY	1011	1011	1011	1011	1011	1011	0
DIAB_A1C_TEST	330	330	330	330	330	330	178
DIAB_BLOOD_SUGAR	531	531	531	531	531	531	178
DIAB_EYE_EXAM	486	486	486	486	486	486	0
DIAB_NEPHRO_SCREEN	498	498	498	498	498	498	0
HTN_PCP_FOLLOWUP	622	622	622	622	622	622	0

Does the check work? Known faults were injected into clean synthetic closures and the checker run on them (recall = share flagged; the last row is the false-positive rate on untouched rows). This tests the check, not InComm's data.

Fault	Injected	Detected	Recall
missing_claim	150	150	1.0000
wrong_member	150	150	1.0000
wrong_cpt	150	150	1.0000
shifted_date	150	150	1.0000
out_of_window	150	150	1.0000
claim_reused_across_members	150	150	1.0000
none (clean rows flagged = false positives)	2100	0	0

7. Reward attribution

- Real data: 0 of 4726 closures follow a reward case (claims 2025-10-29 to 2026-05-26; cases created 2026-06-25). Temporal attribution is impossible; offer timestamps are required (pilot design).
- Causal attribution instead: probability of necessity (the share of treated closers whose closure the offer caused). On the research mock the bounds are 0.02 to 1.00 (monotone point 0.02). The complement is the windfall share of payouts.
- Timing test on the research ledger: MEASURED_ON_RESEARCH_LEDGER; median days to claim offered 194 vs 200 not offered (offer accelerates closure: False).

8. RAF arm (separate from Stars; never summed)

A closure moves RAF only through a risk-adjustment-eligible encounter whose diagnosis maps to a payment HCC that was not already coded this year. Value of an encounter opportunity, not a reward condition: rewards must never depend on a diagnosis.

Care Gap Code	Ra Eligible Encounter	Payment Hcc Dx Share	Dominant Condition	Raf Value Usd Base	Raf Value Upper Bound Usd	Status
BH_POST_DISCHARGE_7D	Yes	0	none	0	n/a	ZERO: diagnosis is not a payment HCC
COPD_SPIROMETRY	No	1.0000	COPD	0	n/a	ZERO: not a risk-adjustment-eligible encounter
DIAB_A1C_TEST	No	1.0000	diabetes	0	n/a	ZERO: not a risk-adjustment-eligible encounter
DIAB_BLOOD_SUGAR	No	1.0000	diabetes	0	n/a	ZERO: not a risk-adjustment-eligible encounter
DIAB_EYE_EXAM	No	1.0000	diabetes	0	n/a	ZERO: not a risk-adjustment-eligi

						ble encounter
DIAB_NEPHRO_SCREEN	No	1.0000	diabetes	0	n/a	ZERO: not a risk-adjustment-eligible encounter
HTN_PCP_FOLLOWUP	Yes	1.0000	heart failure	n/a	4,811	NOT_AVAILABLE: V28 factor not verified (upper bound shown)

9. Priced ROI under literature-anchored lift (a scenario, not a result)

Published incentive trials give relative risks of closure of 1.42 (screening meta-analysis, 95% CI 1.05 to 1.92), 1.17 (colorectal, 8 RCTs) and no effect for diabetic eye screening (IDEAS RCT). These are an external prior, not InComm lift and not the mock lift. Carried as odds ratios so probabilities stay within 0 to 1 at every gap's baseline. Offer-to-all, pay-on-completion. The favourable-value rows stack every favourable bound and are an upper envelope, not an expectation.

Rr Case	Odds Ratio	Mean Implied Rr	Value Case	Net Total Usd	Cost Total Usd	ROI	Attributable Share Of Closers
base	1.2619	1.1489	base	-24,733	110,752	-0.2233	0.1279
base	1.2619	1.1489	high	31,641	110,752	0.2857	0.1279
base	1.2619	1.1489	low	-54,841	110,752	-0.4952	0.1279
high	1.7317	1.3665	base	86,553	128,563	0.6732	0.2604
high	1.7317	1.3665	high	227,504	128,563	1.7696	0.2604
high	1.7317	1.3665	low	11,261	128,563	0.0876	0.2604
low	1.0000	1.0000	base	-98,224	98,224	-1.0000	6.347e-18
low	1.0000	1.0000	high	-98,224	98,224	-1.0000	6.347e-18
low	1.0000	1.0000	low	-98,224	98,224	-1.0000	6.347e-18

Break-even relative risk by gap and value case (the program pays if true RR exceeds it):

Care Gap Code	Value Case	Break Even Rr
BH_POST_DISCHARGE_7D	base	n/a
BH_POST_DISCHARGE_7D	high	n/a

BH_POST_DISCHARGE_7D	low	n/a
COPD_SPIROMETRY	base	n/a
COPD_SPIROMETRY	high	n/a
COPD_SPIROMETRY	low	n/a
DIAB_A1C_TEST	base	1.0826
DIAB_A1C_TEST	high	1.0482
DIAB_A1C_TEST	low	1.1369
DIAB_BLOOD_SUGAR	base	1.0826
DIAB_BLOOD_SUGAR	high	1.0482
DIAB_BLOOD_SUGAR	low	1.1369
DIAB_EYE_EXAM	base	1.1794
DIAB_EYE_EXAM	high	1.1004
DIAB_EYE_EXAM	low	1.3032
DIAB_NEPHRO_SCREEN	base	1.3216
DIAB_NEPHRO_SCREEN	high	1.1607
DIAB_NEPHRO_SCREEN	low	1.6556
HTN_PCP_FOLLOWUP	base	1.1021
HTN_PCP_FOLLOWUP	high	1.0618
HTN_PCP_FOLLOWUP	low	1.1512

9b. Avoided medical cost arm (separate; Stars-only vs Stars plus avoided cost)

Only BH 7-day follow-up has a published closure-to-readmission link (observational; the low case is zero effect). Other gaps are NOT_MODELLED, not zero benefit: screening often raises near-term spending, so no value is guessed. One 30-day event, no discounting; the plan bears the cost under capitation. Under the base literature lift.

Care Gap Code	Value Case	Closure Value Usd	Avoided Cost Usd	Net Per Offer Usd	Net Per Offer With Avoided Usd	Break Even Rr	Break Even Rr With Avoided
BH_POST_DISCHARGE_7D	low	0	1.221e-13	-17.4528	-17.4528	n/a	n/a
BH_POST_DISCHARGE_7D	base	0	198.3673	-17.4528	-9.1981	n/a	1.1259
BH_POST_DISCHARGE_7D	high	0	853.2407	-17.4528	18.0534	n/a	1.0267

9c. Plan-supplied eligible share (what the plan's own mix does to the value)

Eligible share is the share of a plan's enrollees in a measure's denominator. The public anchors (diabetes 26%) may understate an SNP-heavy plan. Set plan_eligible_share in config to your own share. A larger share dilutes each closure, so value falls and the break-even lift rises by the scale shown (multiply the base break-even).

Group	Eligible Share	Mean Usd Per Closure	Break Even Scale
diabetes	0.2000	194.0454	0.8098
diabetes	0.2600	157.1416	1.0000
diabetes	0.3200	134.0768	1.1720
diabetes	0.4000	114.0873	1.3774
diabetes	0.5000	98.0956	1.6019

9d. Value by contract star position (the decision a plan faces)

USD per closure at base inputs, conditional on where a contract sits. Only a 3.5-star contract can be lifted over the 4-star QBP boundary here, so value is concentrated: contracts rated 3.0 or less, 4.0 or more, or 3.5 with the measure far from its next cut point get $0 from this route. Not modelled: a 4.0-star contract near its lower edge protecting its bonus (downside), which would add value for that position. The raw rating inside a half-star band is not published, so it is assumed uniform. The all_contracts row is the portfolio average used in the headline.

Position	N Contracts	Diab Eye Exam	Diab Nephro Screen	Diab A1c Test	Diab Blood Sugar	Htn Pcp Followup	Mean Over Gap Mix
all_contracts	523	143.9000	111.2000	303.0000	303.0000	238.9000	157.1000
rated_3.0_or_less	120	0	0	0	0	0	0
rated_3.5	170	442.8000	341.4000	932.1000	932.1000	713.9000	480.3000
rated_3.5_measure_near_cut	65	1,327	1,383	1,802	1,802	1,447	1,108
rated_3.5_measure_far_from_cut	138	0	0	0	0	0	0
rated_4.0_or_more	233	0	0	0	0	0	0

10. Assumptions and their provenance

Name	Low	Base	High	Unit	Provenance	Source
qbp_per_enrollee_year	318	381	466	USD/enrollee/year	Public source (CMS/KFF)	KFF, MA Quality Bonus Program 2026 (https://www.kff.org/medicare/medicare-will-spend-more-than-13-billion-on-the-medicare-advantage-quality-bonus-program-in-2026/)
eligible_share_diabetes	0.2000	0.2600	0.3200	share of enrollees	Public source (CMS/KFF)	CMS Medicare Beneficiaries at a Glance (26% claims-based); CMS MCBS 2023/2024 (32% self-reported)
eligible_share_hypertension	0.5000	0.6000	0.6500	share of enrollees	Public source (CMS/KFF)	CMS Medicare Beneficiaries at a Glance (65% claims-based); lower bound for the narrower HEDIS denominator

relative_risk_of_closure_with_incentive	1.0000	1.1700	1.4200	RR (incentive vs none)	Published evidence (external; not InComm lift, not mock lift)	Cancer-screening incentive meta-analyses (RR 1.42, 95% CI 1.05-1.92; CRC 30%->35%); IDEAS RCT diabetic eye screening (no benefit). See comment above.
ma_base_payment_pmpm	900	1100	1300	USD per member per month at risk score 1.0	Research assumption (not InComm financials)	County benchmarks are public and vary; not looked up here, so swept
p_condition_not_already_captured_this_year	0.1000	0.3000	0.6000	probability	Research assumption (not InComm financials)	HCCs must be recaptured every calendar year; share of members whose condition is not yet coded this year
bh_30d_readmission_rate_without_followup	0.1100	0.2000	0.3600	probability	Public source (CMS/KFF)	CMS IPF READM-30-IPF mean 20%, facility range 11%-36%

						(doi 10.1176/appi.ps.201900360)
bh_readmission_odds_reduction_from_followup	0	0.1000	0.2200	relative odds reduction	Published evidence (external; not InComm lift, not mock lift)	Marcus 2017 AOR 0.88/0.91 (observational); meta-analysis RRR 0.78 (medical, upper envelope); low = no causal effect
bh_cost_per_readmission	8800	12150	15500	USD per readmission	Public source (CMS/KFF)	HCUP SB189 (Medicare mental/behavioral, 2012) to SB278 (Medicare all-cause, 2018); base = midpoint
rho_C11_boundary_crossing_per_unit_rate	0.0947	0.0982	0.1030	P(tip a 3.5-star contract to 4.0) per unit rate; CMS weight 1	Public source (CMS/KFF) (computed from CMS 2026 Star Ratings data)	523 MA contracts, CMS 2026 cut points and weights
rho_C12_boundary_crossing_per_unit_rate	0.4523	0.4843	0.5049	P(tip a 3.5-star contract to 4.0) per unit rate; CMS weight 3	Public source (CMS/KFF) (computed from CMS 2026 Star	523 MA contracts, CMS 2026 cut points and weights

					Ratings data)	
rho_C13_boundary_crossing_per_unit_rate	0.0687	0.0759	0.0839	P(tip a 3.5-star contract to 4.0) per unit rate; CMS weight 1	Public source (CMS/KFF) (computed from CMS 2026 Star Ratings data)	522 MA contracts, CMS 2026 cut points and weights
rho_C14_boundary_crossing_per_unit_rate	0.4355	0.4757	0.5159	P(tip a 3.5-star contract to 4.0) per unit rate; CMS weight 3	Public source (CMS/KFF) (computed from CMS 2026 Star Ratings data)	508 MA contracts, CMS 2026 cut points and weights

InComm's StarsWeight metadata vs the CMS 2026 weight of the Star measure each gap actually feeds. The model uses CMS's weights; InComm's differ for some measures and several gaps feed no 2026 Part C measure.

Caregapcode	Hedismeasureid	Starsweight	Cms Measure ID	Cms 2026 Weight	Note
BH_POST_DISCHARGE_7D	FUH-7	3		n/a	not a 2026 Part C Star measure
COPD_SPIROMETRY	SPR	1		n/a	not a 2026 Part C Star measure
DIAB_A1C_TEST	CDC-A1C	3	C12	3.0000	matches
DIAB_BLOOD_SUGAR	CDC-A1C	3	C12	3.0000	matches
DIAB_EYE_EXAM	CDC-EYE	3	C11	1.0000	InComm weight differs from CMS
DIAB_NEPHRO_SCREEN	KED	3	C13	1.0000	InComm weight differs from CMS
HTN_PCP_FOLLOWUP	CBP	3	C14	3.0000	matches

11. What remains unknown and why

- Measured lift and a measured dollar ROI: need randomized offers (pilot). The literature-anchored table is a scenario, not a result.
- Temporal reward attribution on real data: every closing claim precedes its reward case in the export; offer timestamps must be recorded before follow-up.
- RAF for heart failure, COPD, depression and non-stage-4 CKD: V28 factors not verified here; an upper bound from the largest verified factor is shown.
- InComm QBP/Star/cost figures: none provided; no InComm financial appears in this document.